



Adaptive Radiotherapy for Reirradiation

Challenges, Recommendations, and Practical Considerations

Rachel Ger
Mass General Brigham Cancer Institute

AMERICAN ASSOCIATION OF PHYSICISTS IN MEDICINE 2026 SUMMER SCHOOL
Transforming Technology, Modern Practice, and Clinical Impact
JUNE 16-20 | UNIVERSITY OF MICHIGAN

1

Disclosures

- None

AMERICAN ASSOCIATION OF PHYSICISTS IN MEDICINE 2026 SUMMER SCHOOL
Transforming Technology, Modern Practice, and Clinical Impact
JUNE 16-20 | UNIVERSITY OF MICHIGAN

2

Clinical Context

- Advances in systemic therapy and improved local control with modern RT techniques have increased survival
- By 2030 over 4 million US survivors will have had prior RT
- Many patients need additional local treatment or for new primary tumors
- Surgery or systemic therapy alone is often not enough

Overestimate Cumulative Dose
Conservative new plan
→ compromised tumor control

Underestimate Cumulative Dose
Aggressive reirradiation
→ serious toxicity or fatal complications

The central challenge: balancing adequate tumor dose against severe late toxicities in previously irradiated tissues, often with incomplete prior dose records.

AMERICAN ASSOCIATION OF PHYSICISTS IN MEDICINE 2026 SUMMER SCHOOL
Transforming Technology, Modern Practice, and Clinical Impact
JUNE 16-20 | UNIVERSITY OF MICHIGAN

3



How ART Helps Address Reirradiation Complexity

- Precise Dose Sculpting**
Minimize OAR exposure while maintaining therapeutic target coverage
- Dynamic Adaptation**
Respond to day-to-day anatomical variation across the treatment course
- Safer Reirradiation**
Enables treatment in anatomically complex or high-risk settings

AMERICAN ASSOCIATION OF PHYSICISTS IN MEDICINE 2026 SUMMER SCHOOL
Transforming Technology, Modern Practice, and Clinical Impact
JUNE 16-20 | UNIVERSITY OF MICHIGAN

4

Three Core Steps in Reirradiation ART Planning



Prior Data **Adaptive Contour** **Cumulative Risk**

AMERICAN ASSOCIATION OF PHYSICISTS IN MEDICINE 2026 SUMMER SCHOOL
Transforming Technology, Modern Practice, and Clinical Impact
JUNE 16-20 | UNIVERSITY OF MICHIGAN

5



Audience Reflection: Your Practice

Question 1
What do you do if the original 3D dose distribution is unavailable?

AMERICAN ASSOCIATION OF PHYSICISTS IN MEDICINE 2026 SUMMER SCHOOL
Transforming Technology, Modern Practice, and Clinical Impact
JUNE 16-20 | UNIVERSITY OF MICHIGAN

6

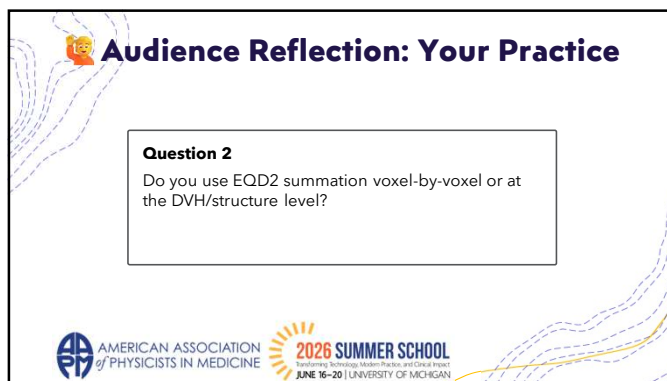


Audience Reflection: Your Practice

AMERICAN ASSOCIATION
of PHYSICISTS IN MEDICINE

2026 SUMMER SCHOOL
Transforming Technology, Modern Practice, and Clinical Impact
JUNE 16-20 | UNIVERSITY OF MICHIGAN

7



Audience Reflection: Your Practice

Question 2

Do you use EQD2 summation voxel-by-voxel or at the DVH/structure level?

AMERICAN ASSOCIATION
of PHYSICISTS IN MEDICINE

2026 SUMMER SCHOOL
Transforming Technology, Modern Practice, and Clinical Impact
JUNE 16-20 | UNIVERSITY OF MICHIGAN

8

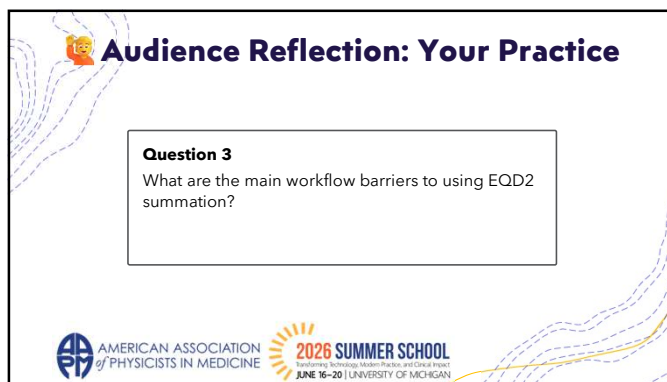


Audience Reflection: Your Practice

AMERICAN ASSOCIATION
of PHYSICISTS IN MEDICINE

2026 SUMMER SCHOOL
Transforming Technology, Modern Practice, and Clinical Impact
JUNE 16-20 | UNIVERSITY OF MICHIGAN

9



Audience Reflection: Your Practice

Question 3

What are the main workflow barriers to using EQD2 summation?

AMERICAN ASSOCIATION
of PHYSICISTS IN MEDICINE

2026 SUMMER SCHOOL
Transforming Technology, Modern Practice, and Clinical Impact
JUNE 16-20 | UNIVERSITY OF MICHIGAN

10

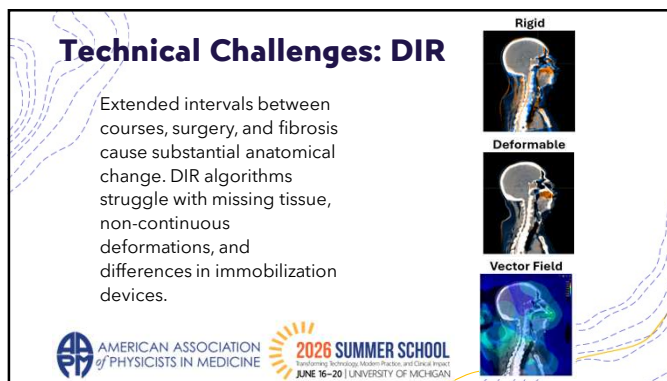


Audience Reflection: Your Practice

AMERICAN ASSOCIATION
of PHYSICISTS IN MEDICINE

2026 SUMMER SCHOOL
Transforming Technology, Modern Practice, and Clinical Impact
JUNE 16-20 | UNIVERSITY OF MICHIGAN

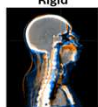
11



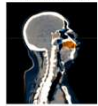
Technical Challenges: DIR

Extended intervals between courses, surgery, and fibrosis cause substantial anatomical change. DIR algorithms struggle with missing tissue, non-continuous deformations, and differences in immobilization devices.

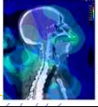
Rigid



Deformable



Vector Field



AMERICAN ASSOCIATION
of PHYSICISTS IN MEDICINE

2026 SUMMER SCHOOL
Transforming Technology, Modern Practice, and Clinical Impact
JUNE 16-20 | UNIVERSITY OF MICHIGAN

12

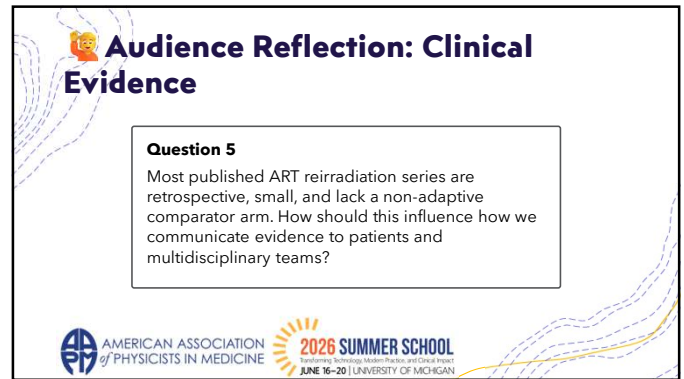


Audience Reflection: Your Practice

AMERICAN ASSOCIATION
of PHYSICISTS IN MEDICINE

2026 SUMMER SCHOOL
Transforming Technology, Modern Practice, and Clinical Impact
JUNE 16-20 | UNIVERSITY OF MICHIGAN

19



Audience Reflection: Clinical Evidence

Question 5

Most published ART reirradiation series are retrospective, small, and lack a non-adaptive comparator arm. How should this influence how we communicate evidence to patients and multidisciplinary teams?

AMERICAN ASSOCIATION
of PHYSICISTS IN MEDICINE

2026 SUMMER SCHOOL
Transforming Technology, Modern Practice, and Clinical Impact
JUNE 16-20 | UNIVERSITY OF MICHIGAN

20



Audience Reflection: Clinical Evidence

AMERICAN ASSOCIATION
of PHYSICISTS IN MEDICINE

2026 SUMMER SCHOOL
Transforming Technology, Modern Practice, and Clinical Impact
JUNE 16-20 | UNIVERSITY OF MICHIGAN

21



Summary Recommendations & Future Directions

Retrieve Prior Plans Always recover historical dose distributions if available; apply proper registration and dose accumulation with rigorous QA and documented uncertainty bounds.	Adapt When It Matters Prioritize ART in cases with significant expected tumor or OAR changes. Online ART for acute daily variation; offline for predictable anatomical drift.
Consider Proton-Based ART Proton ART holds particular promise for reducing integral dose and achieving superior sparing of critical previously irradiated structures.	Build the Evidence Base Support prospective registries (ReCOG, E2-RADIATE/ReCare) and multicenter trials. AI-driven autosegmentation and dose accumulation tools are emerging as critical enablers.

With continued advances in DIR accuracy, radiobiological modeling, imaging integration, and prospective data collection, data collection, adaptive reirradiation is poised to become a standard of care for appropriately selected patients

AMERICAN ASSOCIATION
of PHYSICISTS IN MEDICINE

2026 SUMMER SCHOOL
Transforming Technology, Modern Practice, and Clinical Impact
JUNE 16-20 | UNIVERSITY OF MICHIGAN

22



Acknowledgements



Ryan Oglesby, PhD



Harkiran Kooner, PhD



Cecilia de Sousa, MD, MPH

Contact:
rger@bwh.harvard.edu

AMERICAN ASSOCIATION
of PHYSICISTS IN MEDICINE

2026 SUMMER SCHOOL
Transforming Technology, Modern Practice, and Clinical Impact
JUNE 16-20 | UNIVERSITY OF MICHIGAN

23